

CALCULUS I 2017 Spring Midterm Exam	Dept. or School		proctor	
	Student ID		Name	
※ Your answer must be provided with descriptions how to get the answer. 1. Inverse tangent function, denote by $\tan^{-1}x$, is defined to be the inverse of the restricted tangent function $\left(\tan x, -\frac{\pi}{2} < x < \frac{\pi}{2}\right)$. Find the exact value of the following expressions. (1) (2 points) $\sin(2 \tan^{-1}(-\sqrt{2}))$. (2) (2 points) $\cos(\tan^{-1}2 + \tan^{-1}3)$.	2. (5 points) Find the tangent line approximation $L(x)$ of $f(x)$ at $x = 0$ when $f(x)$ is defined as $f(x) e^{\tan^{-1}x} = \ln \sqrt{1-x^2} + 2\{f(x)\}^2 - 1$, $f(0) > 0$.			

3. We use Newton's method to find an approximated solution to the equation $x^2 - 3 = 0$. Answer the following questions.

(1) (2 points) Start with $x_1 = 1$, find x_2 and x_3 .

(2) (3 points) Find all possible values of x_n , if $x_{n+1} = 3$ for some positive integer n .

4. (6 points) Using the addition formula for tangent, verify the equality $4 \tan^{-1}\left(\frac{1}{5}\right) - \tan^{-1}\left(\frac{1}{239}\right) = \frac{\pi}{4}$.

[illegible]

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7. (5 points) Evaluate the integral $\int \frac{1}{e^{\frac{x}{2}} - e^{\frac{x}{3}}} dx$.

8. (5 points) Evaluate the integral $\int_0^{\sqrt{2}} x^3 \sqrt{9 - 4x^2} dx$.